

A Relighting Method for Single Terrain Image based on Two-stage Albedo Estimation Model

Shun Tatsukawa

Hosei University

Japan

shun.tatsukawa.8x@stu.hosei.ac.jp

Syuhei Sato

Hosei University

Japan

Prometech CG Research

Japan

ssato@hosei.ac.jp

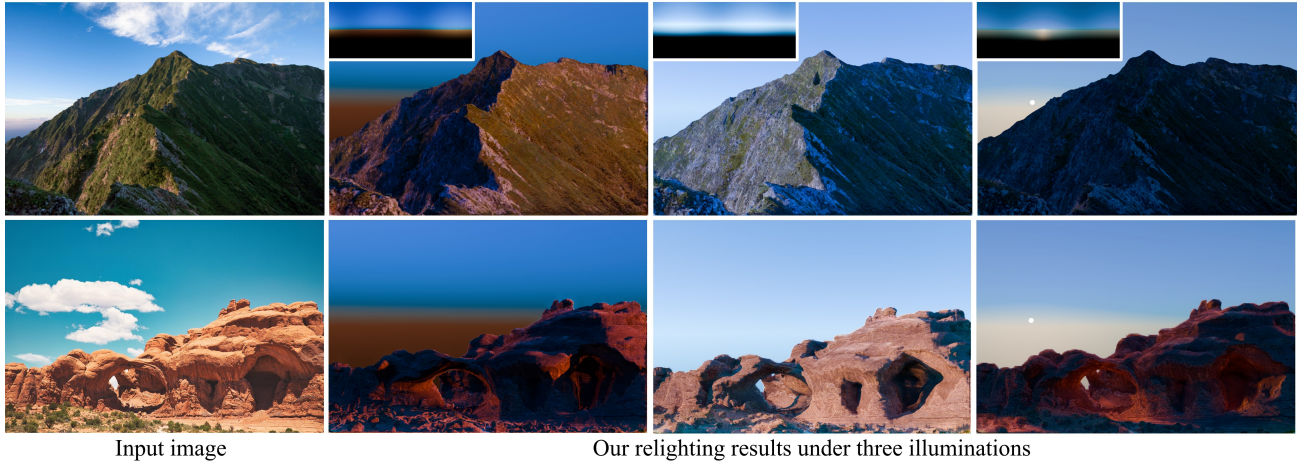


Figure 1: Results created using our relighting system. The input images are sourced from <https://www.pakutaso.com/>.

Abstract

This paper proposes a method for relighting a single terrain image to match user-specified times of day or weather conditions by estimating albedo and depth using deep learning. By employing a two-stage network to remove fog and lighting effects, our method enables accurate albedo estimation even for terrain images containing fog, which has been difficult for previous methods. We validate the effectiveness of our approach by demonstrating several examples and comparisons with an existing method.

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1 Introduction

This study aims to develop a method for relighting large-scale terrain images under desired time-of-day or weather conditions

using deep learning. While several deep learning-based methods for relighting outdoor scenes have been proposed, they are not specifically designed for terrain, leading to unnatural estimations of distant terrain albedo due to fog or atmospheric light scattering, which can result in unsuccessful relighting. Additionally, in the study by Takahashi et al. [2022], which aims to reconstruct 3D terrain, when inputting scenes that differ from the training data, the original terrain features, particularly the albedo, tend to be lost.

To address these issues, we propose a novel method that adopts a two-stage network to remove fog and lighting effects for estimating albedo. We utilize a network architecture suitable for albedo estimation and construct a photorealistic terrain dataset that accounts for atmospheric lighting. These allow us for accurate albedo estimation while preserving the original texture information. We then estimate depth information by another network to generate a point cloud, and relighting is performed using the point cloud combined with the estimated albedo map. Fig. 1 shows examples of relighting using our model. Ours enables the easy generation of terrain images with different lighting conditions.

2 Our Relighting System

Overview of our system is shown in Fig. 2. The masked terrain image is fed into a two-stage composite network designed to remove fog and lighting effects for accurate albedo estimation, while simultaneously being input into a network for depth estimation. The estimated albedo map and depth map are then converted into a

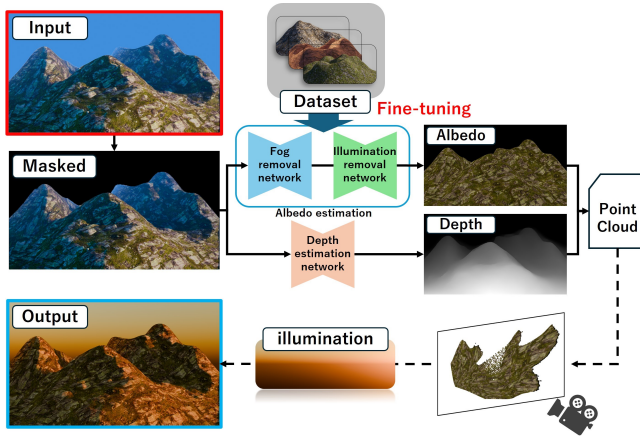


Figure 2: Overview of our relighting model.

point cloud, which is used for relighting. The details of each process are described below.

2.1 Constructing terrain dataset

To create a photorealistic terrain dataset, we adopted the atmospheric scattering model by Nishita et al. [1993]. By randomly adjusting the sunlight and atmospheric parameters within the model, and combining randomly generated terrains with various textures using noise maps, we procedurally generated a diverse set of terrain image data. The dataset aimed at fog removal consists of 300 pairs of images, with and without fog, along with their masked images. Similarly, the dataset aimed at removing lighting effects includes 300 pairs of images with and without lighting effects, along with their masked images.

2.2 Estimating albedo and depth

For removing both fog and lighting effect, we adopted the network architecture of SpA-Former [Zhang et al. 2023], a GAN-based shadow removal model. The distinctive Transformer network in SpA-Former is capable of removing shadows with high precision by considering long-range correlations. Additionally, being a GAN-based generative model, it can also remove highlights and fog, making it well-suited for albedo estimation. We fine-tuned the model's parameters using our dataset.

For depth estimation, we employed the network architecture of MiDaS [Ranftl et al. 2022]. MiDaS is a Zero-Shot Learning model effective for unknown data, allowing for reasonably accurate inferences without the need for additional fine-tuning.

2.3 Creating point cloud and relighting

After estimating the depth map, it is converted into a point cloud. In relighting large-scale images like terrain, it is crucial to replicate the scattering of light in the atmosphere. Therefore, we render the point cloud data, generated from the estimated albedo and depth, using Nishita's atmospheric scattering model [1993] to achieve photorealistic relighting results.

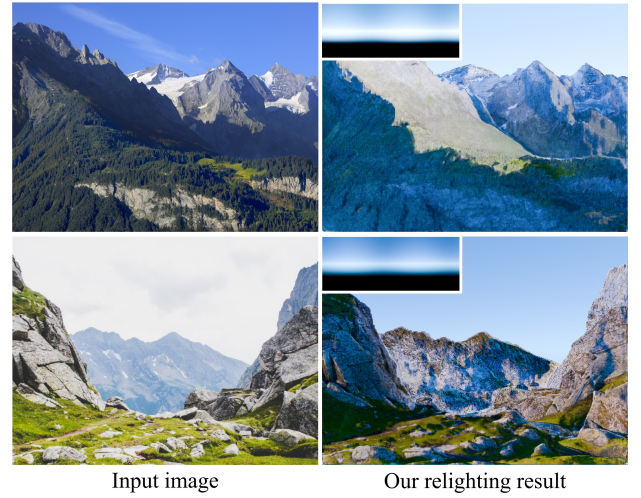


Figure 3: Results for images used in Takahashi et al. [2022].

3 Experimental Results

Fig. 1 shows the results of relighting for real images. Additionally, Fig. 3 is the relighting results for real images used in Takahashi et al. [2022]. The inset image for each result is the environmental map for the relighting. For rendering, we used Blender, a 3DCG software. Additional results, including animations with continuously changing light sources, are included in the supplementary video.

Although a direct comparison is difficult, since Takahashi's method focuses on 3D terrain reconstruction, including hidden surface completion, our method demonstrates that relighting can be achieved while preserving texture details and shapes, at the same viewpoint as the input. It should be noted that in our method, visual artifacts may appear in the shadows because we directly render the point cloud. We plan to explore solutions for this issue, such as generating surfaces from the point clouds, in future work.

4 Conclusions

We proposed a terrain-specific relighting system based on albedo estimation achieved by removing fog and lighting effects. Ours can estimate albedo and depth from real images with higher accuracy compared to Takahashi's method [2022]. As for future work, we plan to develop a method that estimates not only albedo but also specular and diffuse reflectance, as well as atmospheric parameters, to enable relighting across a wider variety of terrain images.

Acknowledgments

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